Chapter 4 exercises

Use real Euclidean inner products unless stated otherwise. Specify the projected subspace, basis, rank assumptions, and residual.

CH04-E1

Let

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

1. Find a basis of $N(A)$.
2. Verify that a row-space basis is orthogonal to the nullspace.
3. Find $N(A^T)$ and explain why $C(A) = \mathbb{R}^2$.

CH04-E2

Let $a = (1, 1, 1)^T$ and $b = (2, 1, 4)^T$.

1. Find the projection p of b onto $L = \text{span}\{a\}$.
2. Find $e = b - p$.
3. Check $a^T e = 0$.
4. Give P and verify $P = P^T = P^2$.

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CH04-E3

Fit $y = c + mt$ to the points

$$(t, y) = (0, 1), (1, 2), (2, 2).$$

1. Give the design matrix A and parameter vector x.
2. Form and solve the normal equations.
3. Find p, e and check $A^T e = 0$.
4. Find the minimum sum of squared residuals.

CH04-E4

Let $A \in \mathbb{R}^{m \times n}$ have independent columns and define

$$P = A(A^T A)^{-1} A^T.$$

Chapter 4 exercises (continued)

1. Prove $P^T = P$ and $P^2 = P$.
2. Prove $I - P$ is a projector and $P(I - P) = 0$.
3. Prove $A^T A$ is invertible.
4. Conversely, prove dependent columns make $A^T A$ singular.

CH04-E5

For

$$a_1 = (1, 1, 0)^T, \qquad a_2 = (1, 0, 1)^T$$

perform Gram-Schmidt.

1. Find orthonormal q_1, q_2.
2. Give R in $A = [a_1\ a_2] = QR$.
3. Check $Q^T Q = I_2$ and $QR = A$.

CH04-E6

Use A from CH04-E5 to solve $Ax = b$ by least squares, with

$$b = (1, 2, 0)^T.$$

1. Compute $Q^T b$ explicitly and solve $Rx = Q^T b$.
2. Find $p = Ax$, $e = b - p$, and $\|e\|^2$.
3. Check $A^T e = 0$ column by column without first assuming the normal equations.

CH04-E7

Let

$$A = \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix}, \qquad b = \begin{bmatrix} 1 \\ 3 \end{bmatrix}.$$

1. Determine whether the columns are independent.
2. Project b onto $C(A)$.
3. Find every least-squares solution $x^\star$.
4. Explain why projection is unique but parameters are not.

CH04-E8

Let

Chapter 4 exercises (continued)

$$P = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

1. Prove $P^2 = P$.
2. Prove $P \neq P^T$.
3. Find the range and kernel of P.
4. Explain why idempotence does not imply orthogonal projection, identifying the failure of orthogonality between the range and kernel.

References and attribution

Gilbert Strang. Introduction to Linear Algebra, 5th Edition, 2016.

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